Reinforcement Learning with Deep-Q Networks on Gymnasium Car Racing Environment

heterogeneous computing - GPUs, exercise 3

October 27 2026

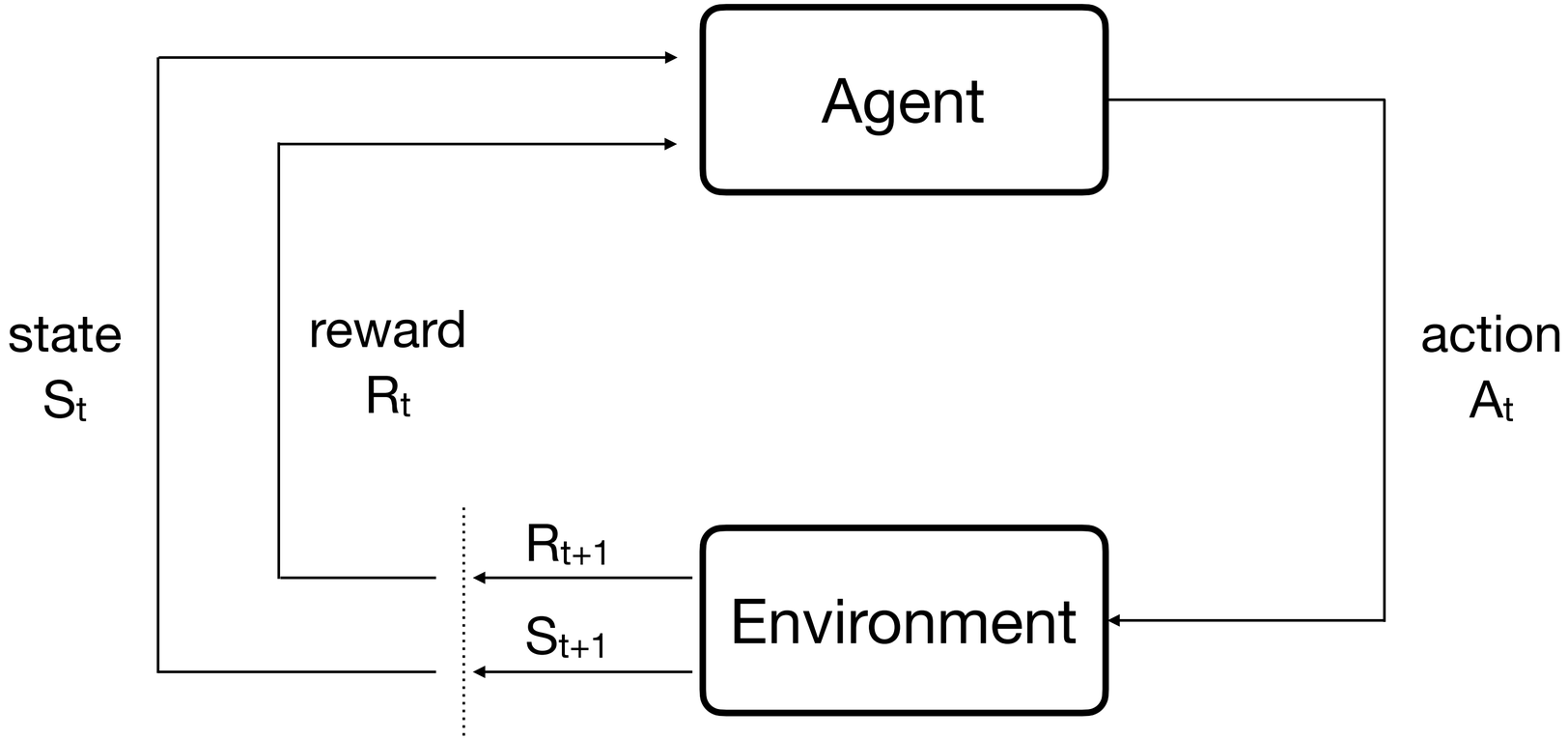
What is Reinforcement Learning?

- Machine Learning Paradigm
- **Applications**: autonomous driving, robot control, gaming, recommendation systems
- In Supervised Learning teaching is done by example (labels)
- In Reinforcement Learning teaching is done by experience (interaction with an environment)
- Addresses the shortcoming of an application where we can’t observe all possibilities of all states

Keywords

- **Policy**: function (neural network) that given a certain state is going to tell which action to be taken
- **Agent**: the entity that learning to make decisions based on an algorithm
- **Environment**: the space the agent interacts with
- **Actions**: a set of possible movements the agent can make
- **State**: the position of the agent in an environment at a given time
- **Reward**: quantifies the value of the predicted action by the policy

Reinforcement Learning Workflow



Reinforcement Learning Workflow

Q-tables

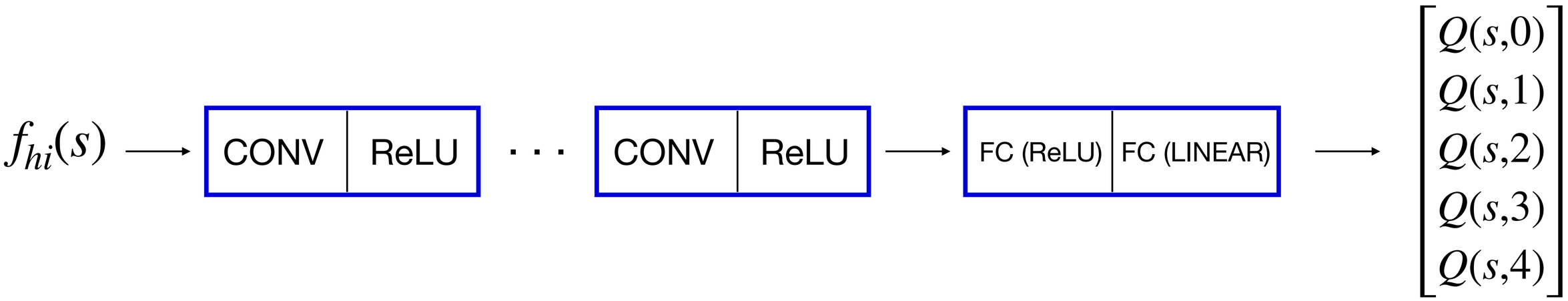
- Simplest form of solving a Reinforcement Learning problem
- Codify how good it is to take an action given a certain state
- * An element of the Q-table *does not contain the immediate reward* for taking an action
    - It follows the Bellman's equation (ideal approximation) using backtracking that determines the optimal action to be taken given a certain state, based on the immediate reward.
    - **Bellman's Equation**: $Q^*(s,a) = R(s) + \gamma \max_{a'} Q^*(s',a')$, where Q* is the expected value in the current state *s* and next state *s'*, and γ is the discount factor (with values between 0 and 1; if value as close to 1 it considers the most recent previous actions in the trajectory, discounting the irrelevant actions.

Q-table example

$$Q = \underbrace{\begin{pmatrix} Q_{11} & Q_{12} \\ Q_{21} & Q_{22} \\ Q_{31} & Q_{32} \\ Q_{41} & Q_{42} \\ Q_{51} & Q_{52} \end{pmatrix}}_{\#\text{actions}} \quad \}\#\text{states}$$

Structure of a Q-table with 2 possible actions and 5 existing states

Policy defined as a CNN for Image Processing



- The fully connected later output size has the same dimension as the action space (e.g., 5).

DQN fundamentals

- Overcomes the memory limitation of Q-tables
- The *policy* is a neural network
- NN in an RL algorithms cannot be trained like in classic supervised learning since they don’t have labels.
    - Uses a model to be trained and a target network
    - The target network is estimated using the Bellman’s equation.
    - A loss function is calculated based on the value of the Bellman’s equation and the immediate reward
    - Calculate the stochastic gradient descent with respect to the weights of the model, and update the weights during backpropagation (refer to the next slide’s diagram)
- *Replay Memory*: used to sample pased experiences during the training process, to limit the corelation between consecutive frames, and limit overfiting.

Deep Q-Networks Implementation Logic

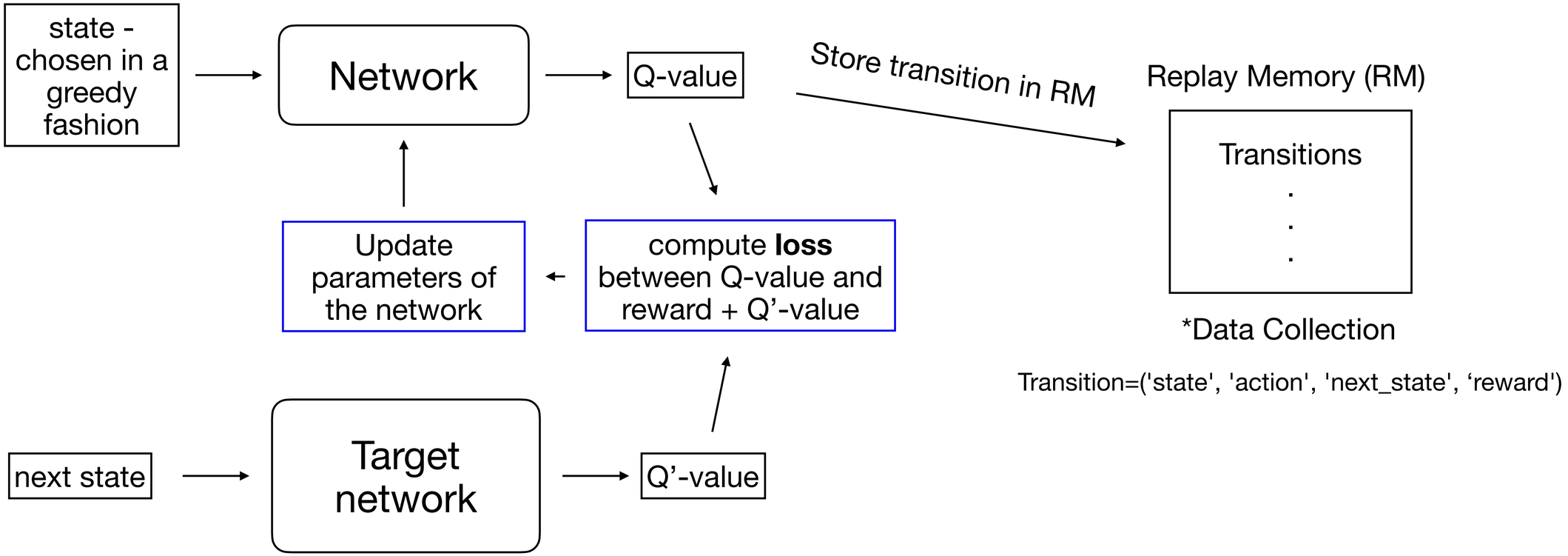


Diagram of the DQN Implementation used in this assignment

Car Racing Environment

- Part of OpenAI's Box2D Gimnasium environments
- It provides 2D 96 x 96 RGB scenarios with a racing car moving along a track
- In the discrete setting (the one we are going to use for this laboratory), the action space provides 5 possibilities: 0 (do nothing), 1 (steer right), 2 (steer left), 3 (gas), 4 (brake)

Car Racing Environment



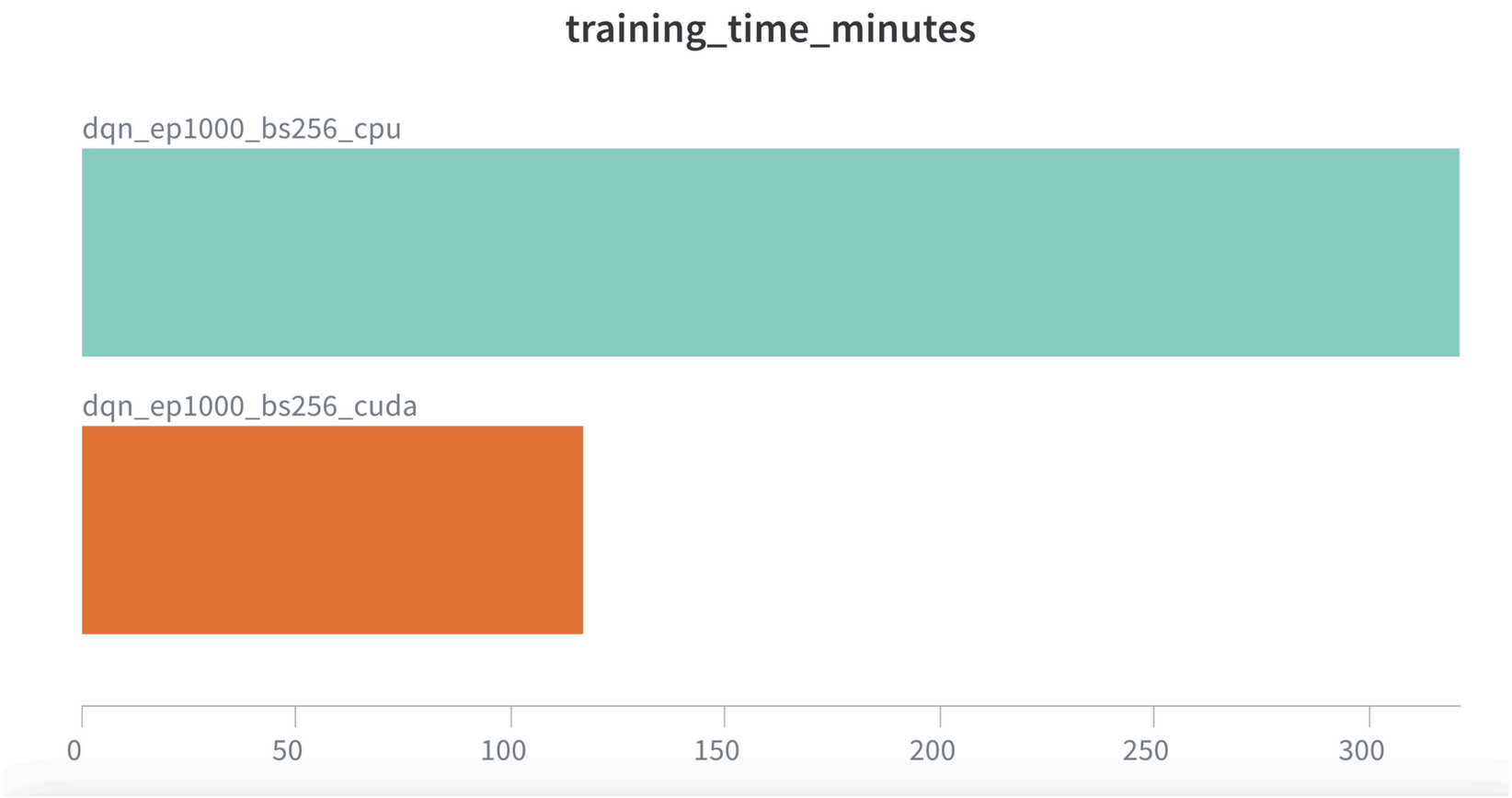
Visualization of the car moving on the track

Limitations of the Car Racing Environment

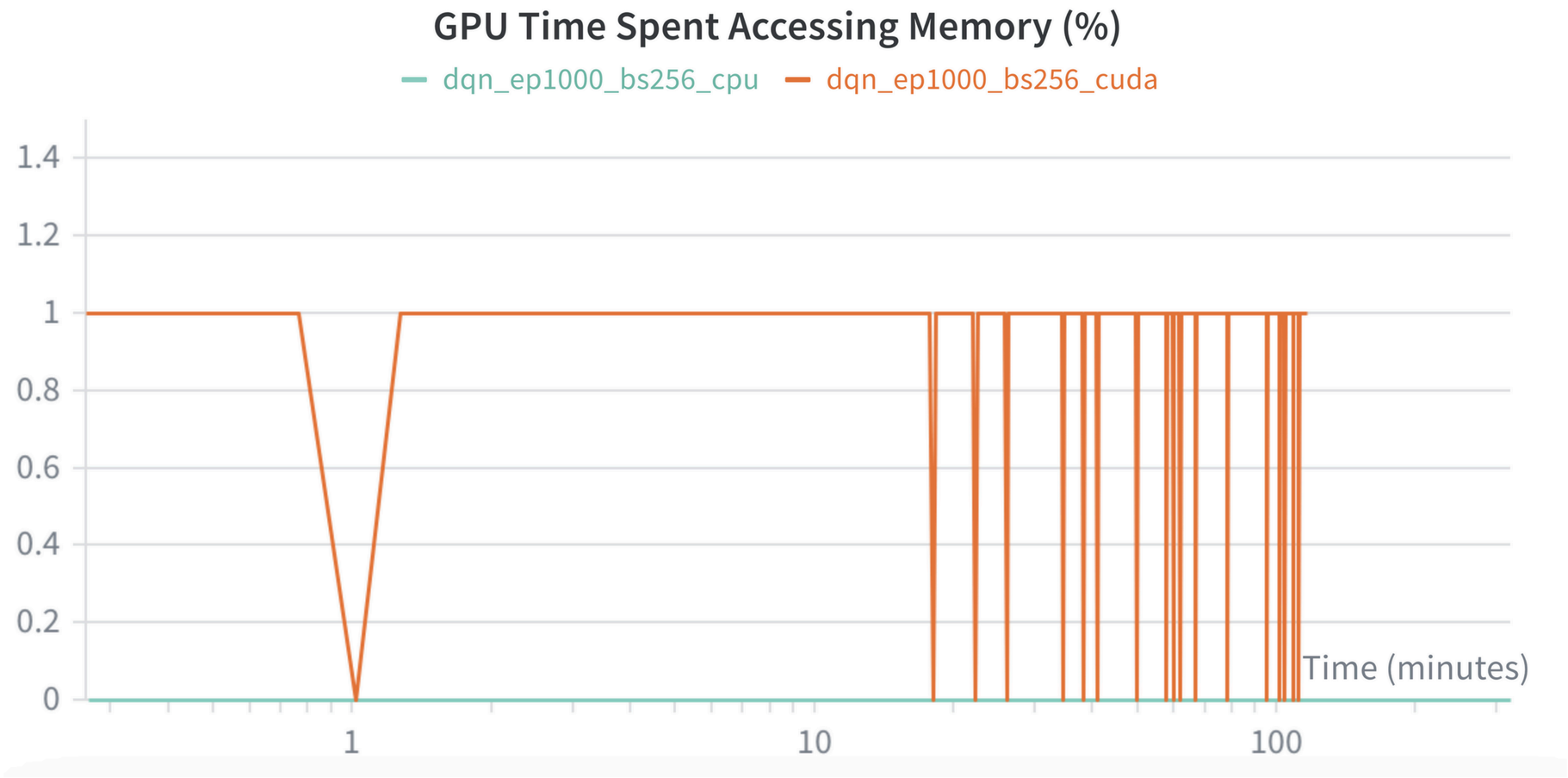
- Heterogeneous lature of the framework
- The rendering/frame collection takes place on the CPU (limitation of the CPU)
- Training takes place on the GPU, if enabled. Here you can take advantage of the complexity of the policy (model to train) to train a better model in less time
- Evaluation takes place on the CPU (reviews assessment)
- The bottleneck is the continuous data movement between the CPU and GPU (try to figure out a way to minimize this limitation

Example Results

https://wandb.ai/narcisaguran20-leibniz-universit-t-hannover/car-racing-dqn?nw=nwusernarcisaguran20

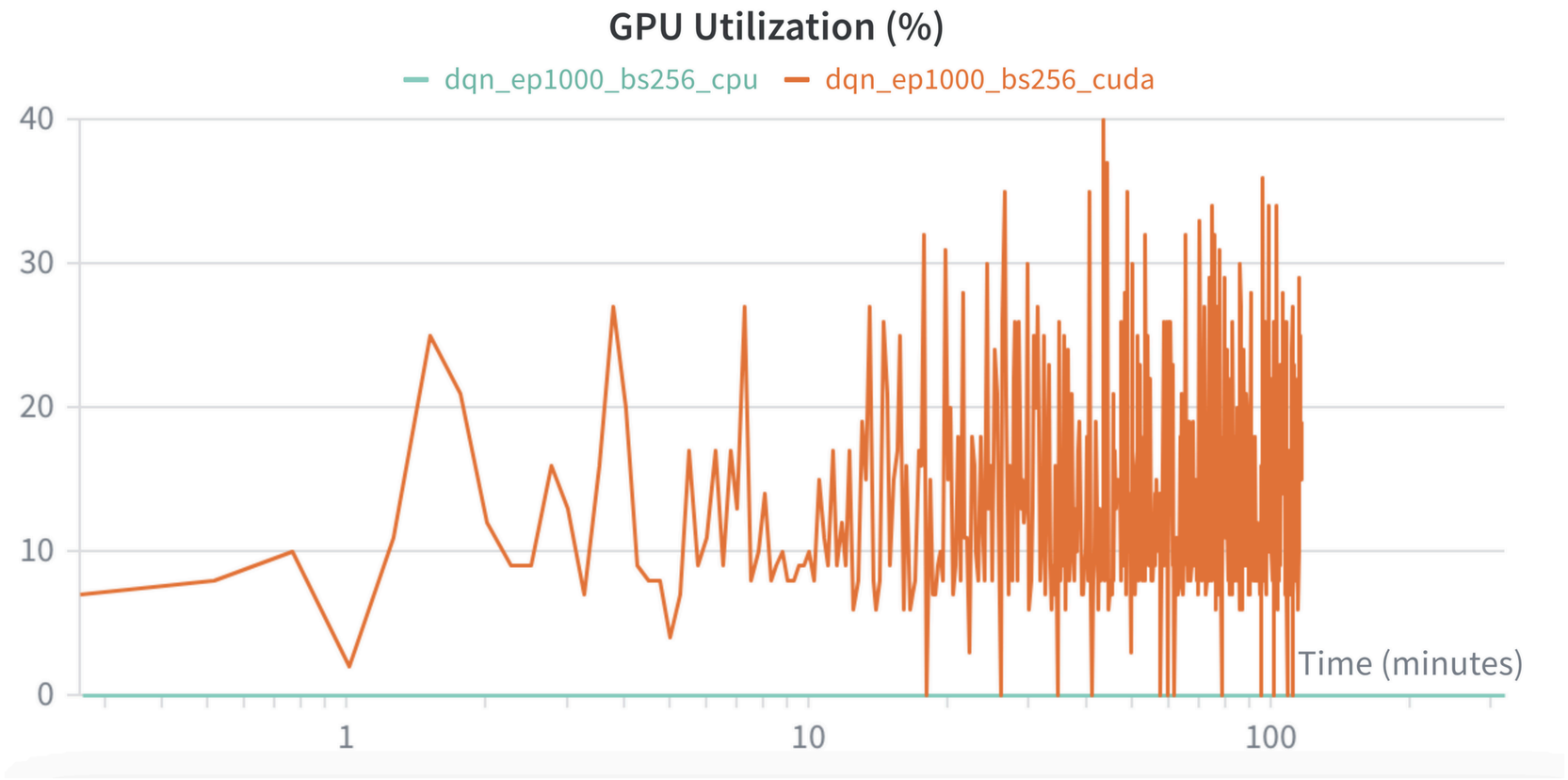


Example Results

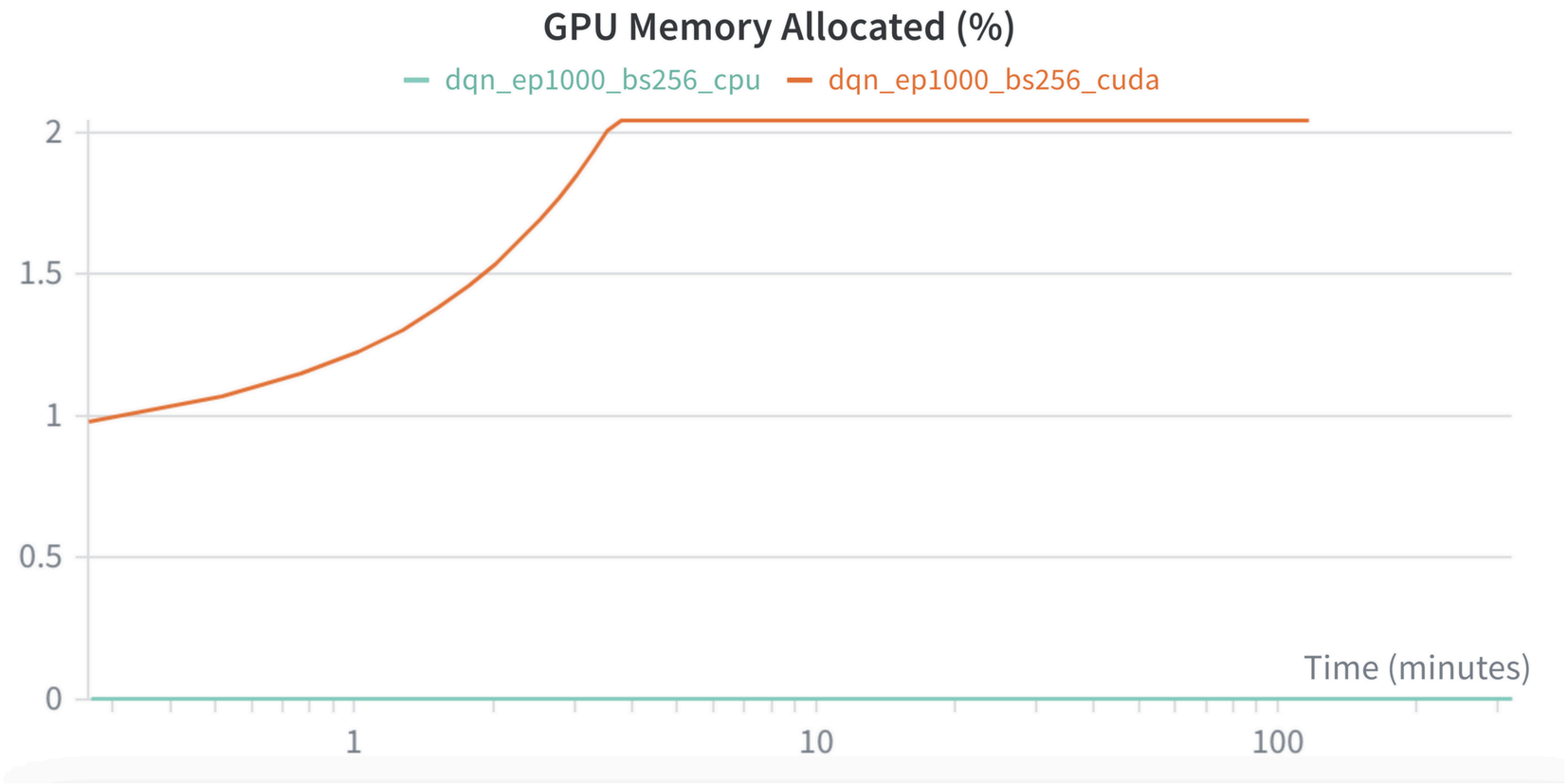


Ideally the time spent accessing memory is 0.

Example Results



Example Results



References:

- Introduction to RL course: https://www.youtube.com/watch?v=4E27qlfYw0A&t=34s
- Seminal Paper DQN: https://www.nature.com/articles/nature14236
- Gymnasium API from OpenAI: https://gymnasium.farama.org/index.html
- Car Racing from Gymnasium: https://gymnasium.farama.org/environments/box2d/car_racing/